

Lesson 8 Control the Robot Arm by the second code

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1. Compile and upload code

Compared with the control code in Lesson 7, the second code has adjusted some functions and joystick control modes.

After power-on, the buzzer will beep once first. Then all servos of each axis will initialize to 90 degrees, and the robotic arm will stand upright. When the claw closes to its limit position, the buzzer will sound three times to remind you to release the corresponding button on the joystick.

A reset function is newly added. Press the triangle button on the joystick, and the robotic arm will slowly return to its standing position.

Most importantly, the second control code allows you to easily set the maximum and minimum operating angles for each servo. You can modify, save and upload the code according to your actual needs. As shown in the figure, lines 23 to 34 of the code define the angles for the 6 servos. It is recommended that the minimum angle of servo6 be no less than 80 degrees, otherwise mechanical interference will occur.

```
21 // ===== Independent Angle Limit for Each Servo ===
22 // Set min/max angle separately to avoid mechanical overtravel
23 #define SERVO1_MIN 0
24 #define SERVO1_MAX 180
25 #define SERVO2_MIN 0
26 #define SERVO2_MAX 180
27 #define SERVO3_MIN 0
28 #define SERVO3_MAX 180
29 #define SERVO4_MIN 0
30 #define SERVO4_MAX 180
31 #define SERVO5_MIN 0
32 #define SERVO5_MAX 180
33 #define SERVO6_MIN 85
34 #define SERVO6_MAX 180
35
```

In addition, compared with the first code of Lesson 7, this version removes two features: RGB light lighting and action recording and doing.

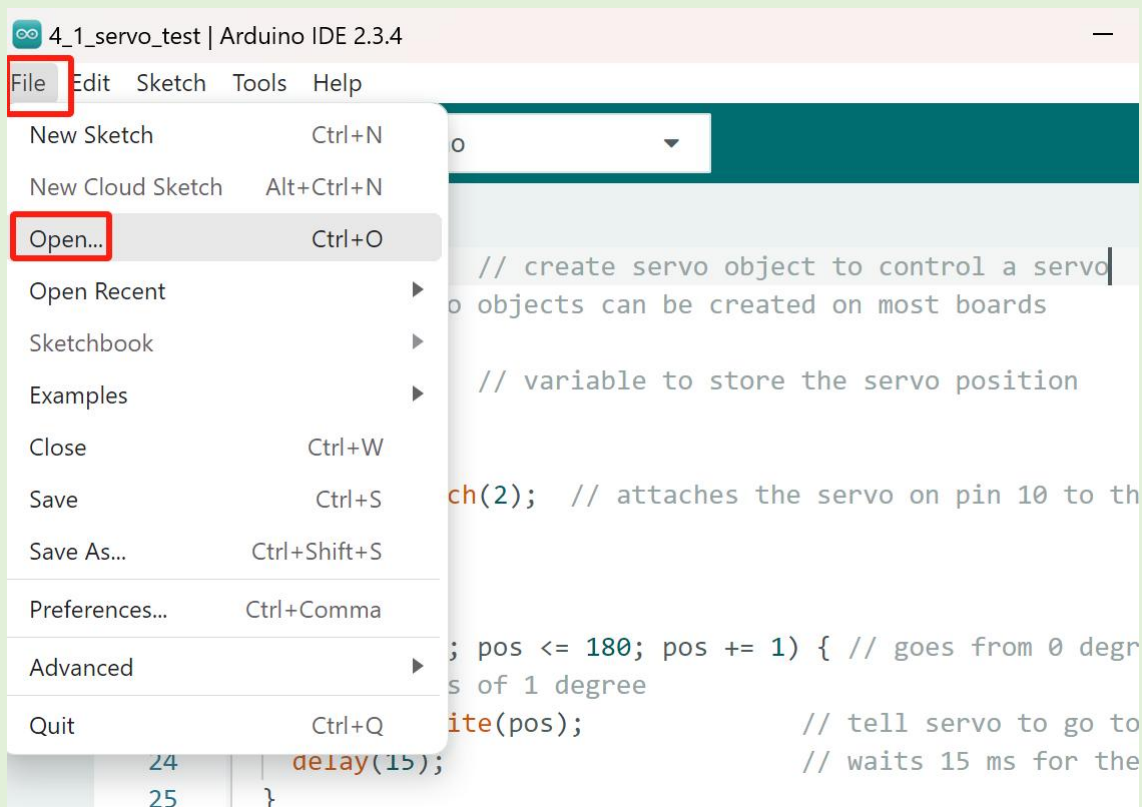
1.1 Open the code

The code used in this lesson is placed in this folder: “<E:\CKK0017-main\Tutorial\sketches>”

1.1.1 Double-click the Arduino IDE shortcut on the desktop to open it

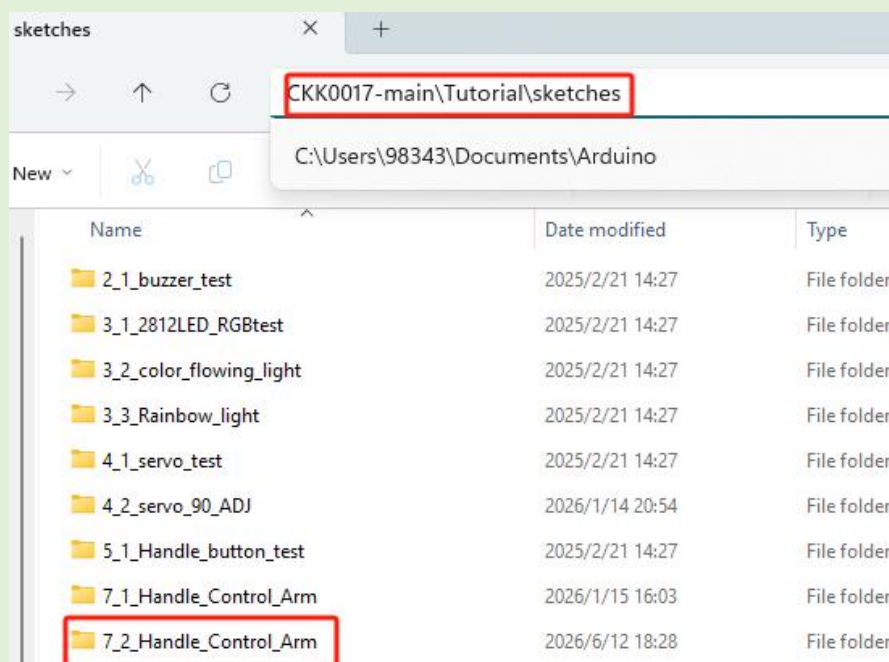


1.1.2 Click "File" --- "open"

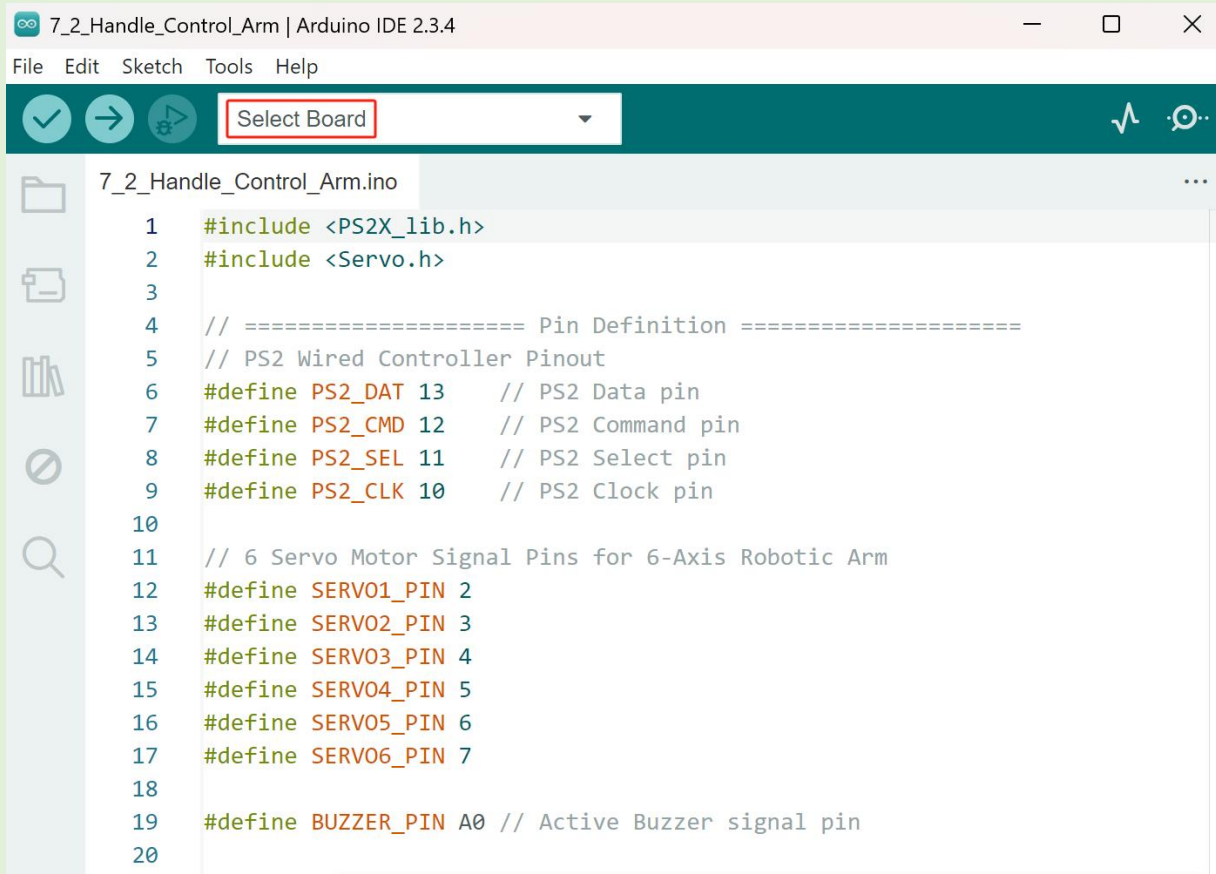


1.1.3 Select the code in the folder named 7_2_Handle_Control_Arm:

E:\CKK0017-main\Tutorial\sketches\7_2_Handle_Control_Arm.

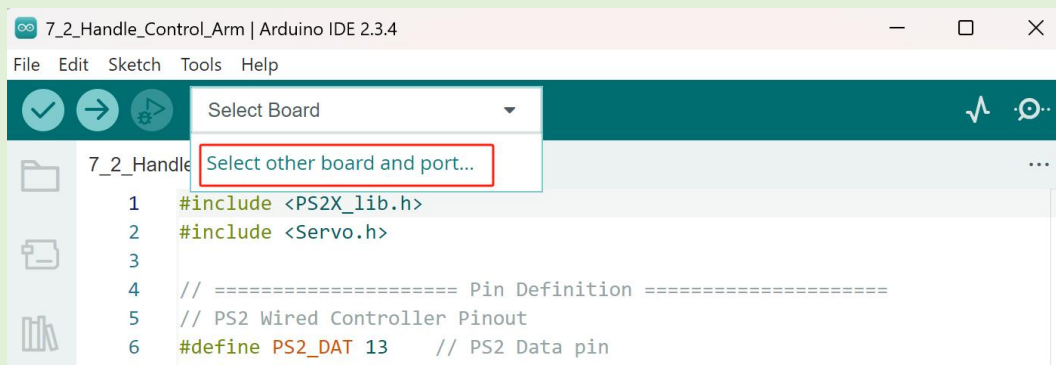


Click "open", Open the code interface as follows:

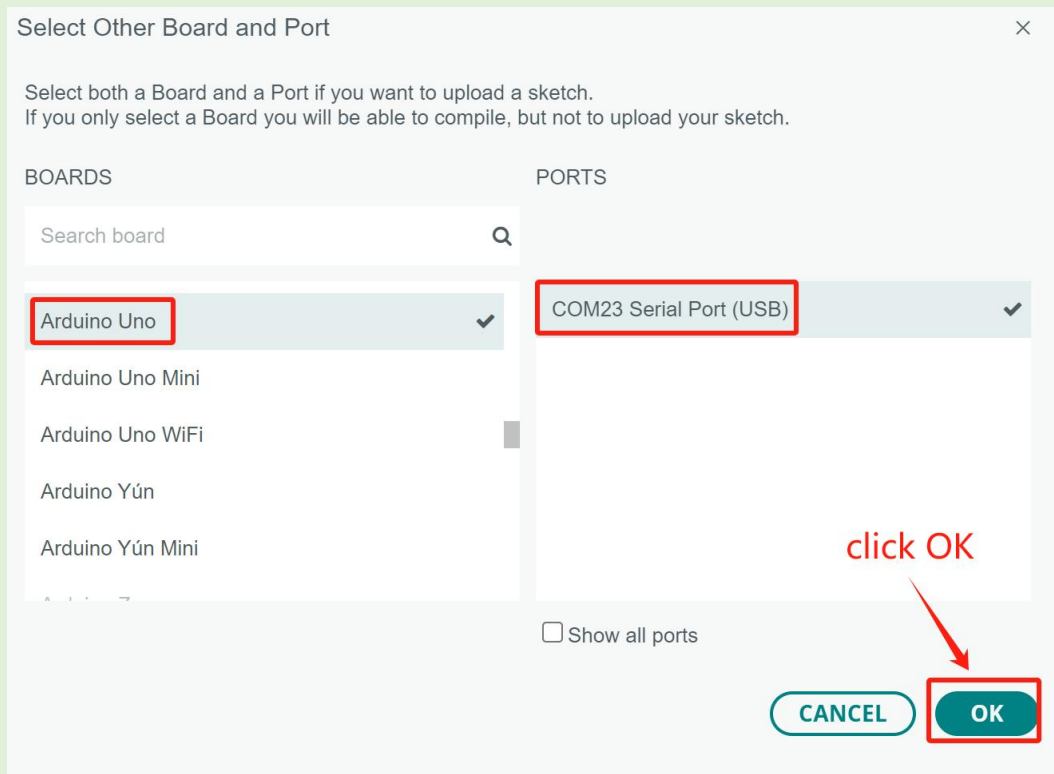


1.2 Select development board and port

1.2.1 Click the “Select Board”, then click “Select other board and port...”



1.2.2 Click and select Arduino UNO in the "BOARDS" dropdown menu, then select the corresponding COM port of Robot Arm Hat on the computer in the "PORTS" menu. On my computer, it recognizes and displays COM23 (COM port is commonly known as an input output port for a device normally PC which enables communication between Arduino and PC. You can check your arduino com number in device manager, the com port of our arduino board is recognized as COM23 in this tutorial), then click “OK”.



1.3 Compile and upload code

1.3.1 Click compile button  , successfully compiled the code will display “Done compiling”

1.3.2 Click upload button  , successfully uploading the code will display “Done uploading”.

2. Safety and Precautions(important)

This metal robotic arm stands at a height of 45cm when fully standing, with a maximum swing arm length of 36cm. Adopting 5 maximum torques of 20KG CM and a maximum torque of 25KG CM's servo motor. Therefore, this metal robotic arm has a relatively large range of travel and rotational force during operation. When it is working, we need to be careful to keep our bodies away from its range of motion and avoid being touched by the moving robotic arm.

Before turning on the power to the control board, first hold the robotic arm upright by hand. This way, after turning on the power, the initialization action of the robotic arm will be relatively small.

When using the Handle controller to control the robotic arm, the body should maintain a distance of at least 0.5 meters from the robotic arm. When the robotic arm is in motion, hands or other parts of the body should not approach the robotic arm, and the body should not touch the joints of the robotic arm to prevent collision or self injury.

When there is a malfunction in the operation of the robotic arm, such as being entangled in joints by wires, tapes, or other similar objects, it is necessary to first turn off the power switch on the robotic arm control board before dealing with obstacles on the robotic arm.

This robotic arm is suitable for teenagers aged 15 and above to learn and use.

Children aged 8-15 need to use this robotic arm under adult supervision.

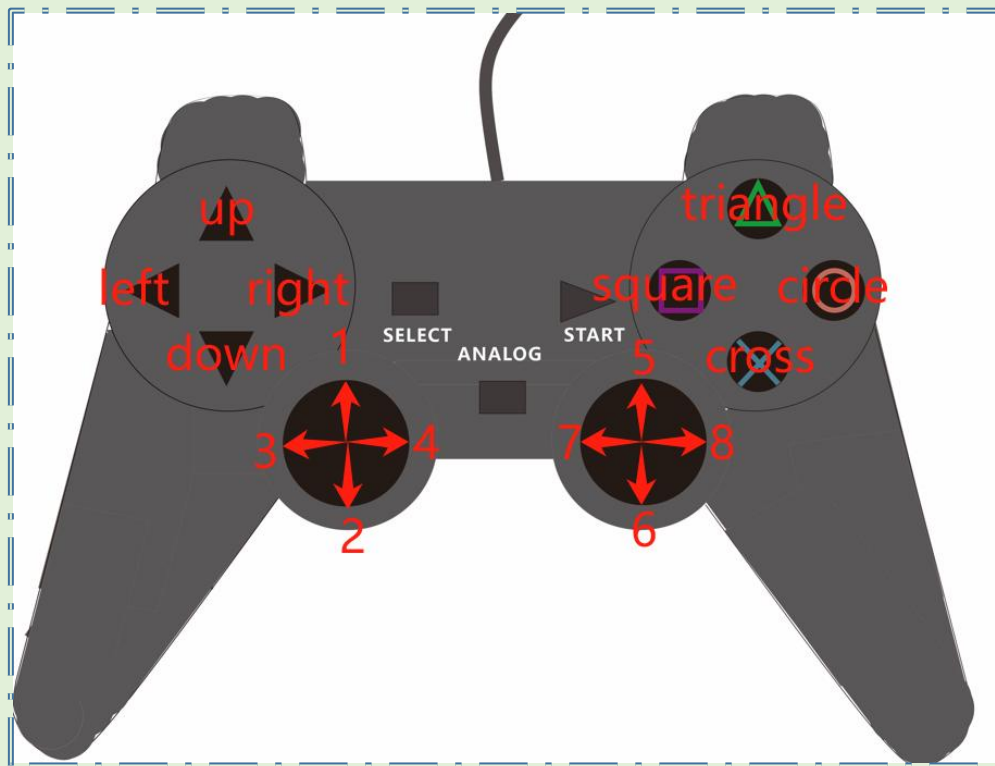
Children under 8 years old are not allowed to use it and it needs to be kept out of their reach.

3. How to control the robotic arm by the Handle controller

There are buttons and joysticks on the Handle controller. We have defined the servos and actions controlled by each button/joystick in the code, so that the robotic arm can be controlled through the Handle controller.

3.1 Understand the buttons and joystick on the Handle controller

The Handle controller has left and right joysticks, each of which is divided into four control directions: front, back, left, and right. For the convenience of distinction, we use the number 1 to represent the front (control direction) of the left joystick, the number 2 to represent the back (control direction) of the left joystick, the number 3 to represent the left (control direction) of the left joystick, the number 4 to represent the right (control direction) of the left joystick, the number 5 to represent the front (control direction) of the right joystick, the number 6 to represent the back (control direction) of the right joystick, the number 7 to represent the left (control direction) of the right joystick, and the number 8 to represent the right (control direction) of the right joystick. The buttons include up, down, left, right, triangle, cross, square, circle, as shown in the following figure:



In this second code, it is defined that the joystick controls the actions of servo2 (6DF-2) in the 1 and 2 directions, servo1 (6DF-1) in the 3 and 4 directions, servo4 (6DF-4) in the 5 and 6 directions, servo3 (6DF-3) in the 7 and 8 directions. servo5 (6DF-5) in the up and down buttons. and servo6 (6DF-6) in the left and right buttons.

4. Trouble Shooting

4.1 Arduino IDE failed to recognize Robot Arm Hat

---Check if the USB driver is installed. To install the CH340 driver, please refer to Lesson 1.

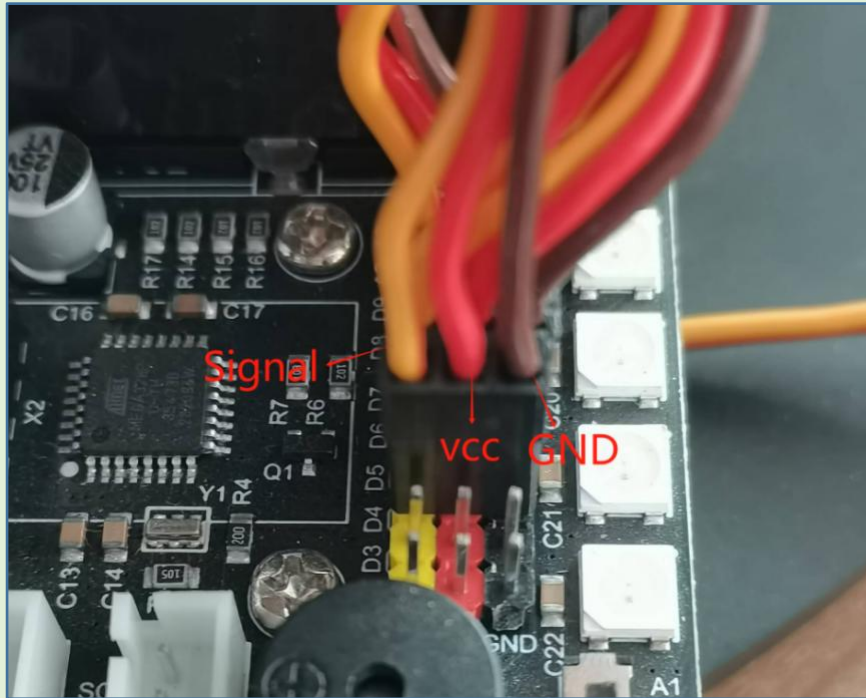
---Check if the USB port of the computer is working properly. You can insert a USB flash drive for testing. If the computer can read the USB flash drive, the USB port is working properly.

---Check if there is a problem with the USB cable. If the USB port of the computer is functioning properly, connect the computer and phone with a USB cable. If the computer can read files from the phone, then the USB cable is functioning properly.

4.2 The robotic arm is not working

---Check if the power light on the Robot Arm Drive Hat is on. If it is not on, it means that there is no power on. Check if the switch is turned to the "ON" state, and if the power adapter is plugged into the Robot Arm Drive Hat and powered on.

---Check if the female terminal of the servo motor is plugged into the correct position on the Robot Arm Drive Hat. Signal corresponds to the yellow pin, VCC corresponds to the red pin, and GND corresponds to the black pin.



4.3 The working methods do not match the description

---Confirm that all 6 servos have been tested and adjusted to 90 degrees before assembly.

---The servos 1, 2, 3, 4, 5, and 6 correspond one-to-one to D2, D3, D4, D5, D6, D7 on the Robot Arm Drive Hat, Confirm that the servo sequence is not connected incorrectly.

---Confirm that the servo cable is long enough and not wrapped around the robotic arm.

4.4 Robotic arm shaking

---The robotic arm shakes without operating the Handle controller. This is because there is a gap in the servo gear, which is around 3° . When the robotic arm is in a standing position (when the center of gravity is not vertically downward), all the weight of the upper part of the Axis2 is loaded onto the Servo2, and under the action of gravity, it drives the servo panel on the Servo2 to rotate downwards

towards the clearance angle. When the servo panel rotates to the limit position of the gear clearance, due to the inertia of gravity, it drives the Servo2 gear to rotate downwards by some angle. However, the program in the Robot Arm Drive Hat needs to maintain the angle before Servo2, and then drive Servo2 to rotate upwards back, and then Servo2 will rotate towards the gear clearance under the action of gravity. This cycle creates shaking.

Solution: Use your hands to straighten the body of the robotic arm so that it stands completely vertically with its center of gravity on Servo2 and vertically downward. At this point, the robotic arm will stop shaking.

5. Any questions and suggestions are welcome

Thank you for reading this document!

If you find any errors and omissions in the tutorial, or if you have any suggestions and questions, please feel free to contact us:

cokoino@outlook.com

We will do our best to make changes and publish revisions as soon as possible.

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